

VIETNAM NATIONAL UNIVERSITY-HO CHI MINH CITY

HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

KNOWLEDGE ENGINEERING DEPARTMENT

Homework 6

Course: Introduction to Artificial Intelligence

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1 Propositional Logic (PL)

1.1 Assignment 1: Transform formulas into conjunctive normal form (CNF)

1.1.1 $A \iff B$

Step 1: Rewrite the biconditional using implication

$$A \iff B \equiv (A \Rightarrow B) \wedge (B \Rightarrow A)$$

Step 2: Replace implications

$$(A \Rightarrow B) \equiv (\neg A \vee B), \quad (B \Rightarrow A) \equiv (\neg B \vee A)$$

so

$$(\neg A \vee B) \wedge (\neg B \vee A)$$

Result (CNF):

$$(\neg A \vee B) \wedge (\neg B \vee A)$$

1.1.2 $A \wedge B \iff A \vee B$

Step 1: Rewrite the biconditional

$$(A \wedge B \Rightarrow A \vee B) \wedge (A \vee B \Rightarrow A \wedge B)$$

Step 2: Replace implications

$$(\neg(A \wedge B) \vee (A \vee B)) \wedge (\neg(A \vee B) \vee (A \wedge B))$$

Step 3: Apply De Morgan's laws

$$(\neg A \vee \neg B \vee A \vee B) \wedge (\neg A \wedge \neg B \vee A \wedge B)$$

- Simplify the first clause:

$$(\neg A \vee A \vee \neg B \vee B) \equiv (\text{always true}) \equiv \top$$

- Second clause:

$$(\neg A \wedge \neg B) \vee (A \wedge B)$$

- Use distributive law to convert to CNF:

$$(\neg A \vee A) \wedge (\neg A \vee B) \wedge (\neg B \vee A) \wedge (\neg B \vee B)$$

- Remove tautologies:

$$(\neg A \vee B) \wedge (\neg B \vee A)$$

Result (CNF):

$$(\neg A \vee B) \wedge (\neg B \vee A)$$

1.1.3 $A \wedge (A \Rightarrow B) \Rightarrow B$ **Step 1:** Eliminate Implications

$$\neg(A \wedge (\neg A \vee B)) \vee B$$

Step 2: Apply De Morgan's Law

- Apply De Morgan:

$$(\neg A \vee \neg(\neg A \vee B)) \vee B$$

- Simplify inner negation:

$$(\neg A \vee (A \wedge \neg B)) \vee B$$

Step 3: Distribute

- Use distributive law:

$$((\neg A \vee A) \wedge (\neg A \vee \neg B)) \vee B$$

- Simplify:

$$(\neg A \vee \neg B) \vee B \equiv \neg A \vee \neg B \vee B \equiv \neg A \vee (\neg B \vee B) \equiv \neg A \vee TRUE$$

Result (CNF): TRUE**1.2 Assignment 2: Prove the sentence C is entailed by KB**

Given a knowledge base as follows:

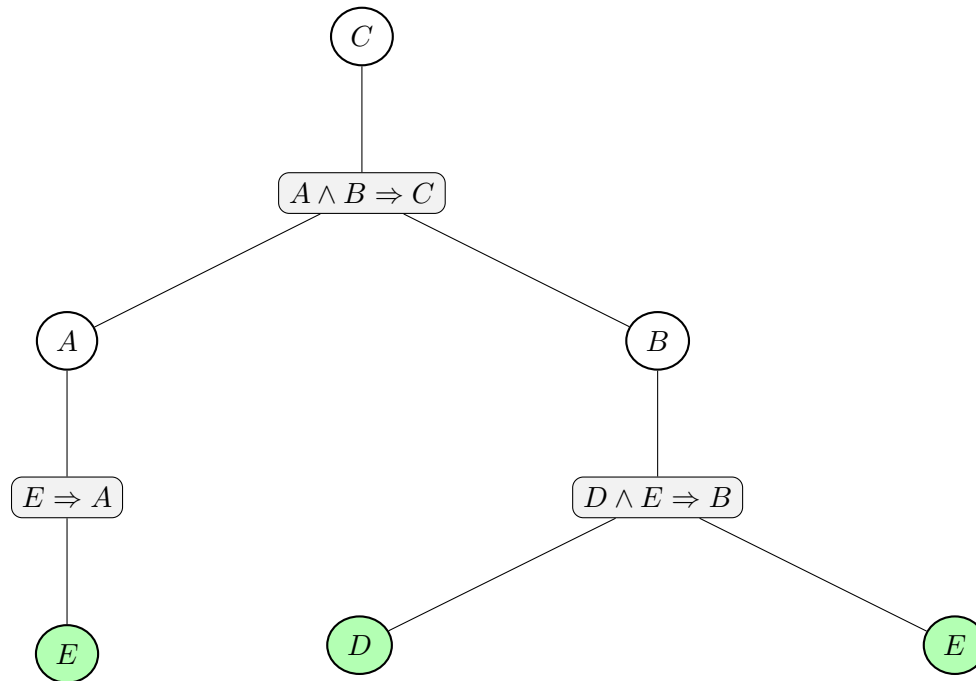
$$KB = \{ A \wedge B \Rightarrow C, D \wedge E \Rightarrow B, F \Rightarrow A, E \Rightarrow A, E, D \}$$

1.2.1 Forward chaining

#	Clause	Explanation
(1)	$A \wedge B \Rightarrow C$	From KB
(2)	$D \wedge E \Rightarrow B$	From KB
(3)	$F \Rightarrow A$	From KB
(4)	$E \Rightarrow A$	From KB
(5)	E	From KB
(6)	D	From KB
(7)	B	From (6), (5), (2)
(8)	A	From (5), (4)
(9)	C	From (8), (7), (1)
Conclusion: C is entailed by KB.		

1.2.2 Backward chaining

#	Clause	Explanation
(1)	$A \wedge B \Rightarrow C$	From KB
(2)	$D \wedge E \Rightarrow B$	From KB
(3)	$F \Rightarrow A$	From KB
(4)	$E \Rightarrow A$	From KB
(5)	E	From KB
(6)	D	From KB



1.3 Assignment 3: Using a resolution inference to proof

Propositions:

- A : The criminal had an accomplice
- C : The criminal came in a car
- K : The criminal had the key

Given statements:

- $A \Rightarrow C$
- $(\neg A \wedge \neg K) \vee (K \wedge A)$
- K

Goal: Is C entailed?

Step 1: Convert to CNF

- $A \Rightarrow C \equiv \neg A \vee C$
- $(\neg A \wedge \neg K) \vee (K \wedge A) \equiv ((\neg A \vee K) \wedge (\neg A \vee A)) \wedge ((\neg K \vee K) \wedge (\neg K \vee A)) \equiv (\neg A \vee K) \wedge (\neg K \vee A)$
- Already in CNF: K
- Negated goal: $\neg C$

CNF Clause Set

1. $\neg A \vee C$: the criminal had no accomplice, or the criminal came in a car
2. $\neg A \vee K$: distributing clause (2)
3. $\neg K \vee A$: distributing clause (2)
4. K : the criminal had the key
5. $\neg C$: negation of goal for refutation

Step 2: Resolution

#	Clause	Explanation
(1)	$\neg A \vee C$	From KB
(2)	$\neg A \vee K$	From KB
(3)	$\neg K \vee A$	From KB
(4)	K	From KB
(5)	$\neg C$	Negation of goal for refutation
(6)	$\neg A$	Resolve (1) and (5): $(\neg A \vee C), (\neg C)$
(7)	$\neg K$	Resolve (3) and (6): $(\neg K \vee A), (\neg A)$
(8)	$\perp$	Contradiction from (4) and (7): $(\neg K), (K)$
Result: Contradiction $\perp$ is reached		

Conclusion: Since a contradiction is reached, C is entailed by the knowledge base. Therefore, **the criminal did come in a car.**

2 First-Order Logic (FOL)**2.1 Assignment 4: Write the FOL sentence to express meaning**

The predicates:

- $CAT(x)$: x is a cat
- $MOUSE(x)$: x is a mouse
- $CHASES(x, y)$: x chases y

English sentences:

1. Every cat chases every mouse.
2. For every cat, there is a mouse that the cat chases.
3. There is a cat who chases every mouse.
4. Some cat chases some mouse.
5. There is a mouse that every cat chases.
6. For every mouse, there is a cat who chases that mouse.

Translate to FOL sentences:

1. $\forall x (CAT(x) \Rightarrow \forall y (MOUSE(y) \Rightarrow CHASES(x, y)))$
2. $\forall x (CAT(x) \Rightarrow \exists y (MOUSE(y) \wedge CHASES(x, y)))$
3. $\exists x (CAT(x) \wedge \forall y (MOUSE(y) \Rightarrow CHASES(x, y)))$
4. $\exists x (CAT(x) \wedge \exists y (MOUSE(y) \wedge CHASES(x, y)))$
5. $\exists y (MOUSE(y) \wedge \forall x (CAT(x) \Rightarrow CHASES(x, y)))$
6. $\forall y (MOUSE(y) \Rightarrow \exists x (CAT(x) \wedge CHASES(x, y)))$

2.2 Assignment 5**English sentences:**

1. Every child loves Santa.
2. Everyone who loves Santa loves any reindeer.
3. Rudolph is a reindeer, and Rudolph has a red nose.
4. Anything which has a red nose is weird or is a clown.
5. No reindeer is a clown.
6. Scrooge does not love anything which is weird.

2.2.1 Build a FOL knowledge

The predicates:

- $Child(x)$: x is a child
- $Loves(x, y)$: x loves y
- $Reindeer(x)$: x is a reindeer
- $RedNose(x)$: x has a red nose
- $Weird(x)$: x is weird
- $Clown(x)$: x is a clown

Translate to FOL sentences:

1. $\forall x (Child(x) \Rightarrow Loves(x, Santa))$
2. $\forall x (Loves(x, Santa) \Rightarrow \forall y (Reindeer(y) \Rightarrow Loves(x, y)))$
3. $Reindeer(Rudolph) \wedge RedNose(Rudolph)$
4. $\forall x (RedNose(x) \Rightarrow Weird(x) \vee Clown(x))$
5. $\forall x (Reindeer(x) \Rightarrow \neg Clown(x))$
6. $\forall x (Weird(x) \Rightarrow \neg Loves(Scrooge, x))$

2.2.2 Convert each FOL sentence in the knowledge base into CNF

1. $\neg Child(x) \vee Loves(x, Santa)$
2. $\neg Loves(x, Santa) \vee \neg Reindeer(y) \vee Loves(x, y)$
3. $Reindeer(Rudolph)$
 $RedNose(Rudolph)$
4. $\neg RedNose(x) \vee Weird(x) \vee Clown(x)$
5. $\neg Reindeer(x) \vee \neg Clown(x)$
6. $\neg Weird(x) \vee \neg Loves(Scrooge, x)$

2.2.3 Use resolution to prove that “Scrooge is not a child.”**Constants:**

- *Santa*: Santa
- *Rudolph*: Rudolph
- *Scrooge*: S

#	Clause	Explanation
(1)	$\neg \text{Child}(x) \vee \text{Loves}(x, \text{Santa})$	From KB
(2)	$\neg \text{Loves}(x, \text{Santa}) \vee \neg \text{Reindeer}(y) \vee \text{Loves}(x, y)$	From KB
(3)	$\text{Reindeer}(\text{Ru})$	From KB
(4)	$\text{RedNose}(\text{Ru})$	From KB
(5)	$\neg \text{RedNose}(x) \vee \text{Weird}(x) \vee \text{Clown}(x)$	From KB
(6)	$\neg \text{Reindeer}(x) \vee \neg \text{Clown}(x)$	From KB
(7)	$\neg \text{Weird}(x) \vee \neg \text{Loves}(S, x)$	From KB
(8)	$\text{Child}(S)$	Negation of goal
(9)	$\text{Loves}(S, \text{Santa})$	From (1) and (8)
(10)	$\neg \text{Loves}(S, \text{Santa}) \vee \neg \text{Reindeer}(\text{Ru}) \vee \text{Loves}(S, \text{Ru})$	From (2) with $x = \text{Scrooge}$, $y = \text{Ru}$
(11)	$\neg \text{Reindeer}(\text{Ru}) \vee \text{Loves}(S, \text{Ru})$	Resolve (10) and (9)
(12)	$\text{Loves}(S, \text{Ru})$	Resolve (11) and (3)
(13)	$\neg \text{RedNose}(\text{Ru}) \vee \text{Weird}(\text{Ru}) \vee \text{Clown}(\text{Ru})$	From (5)
(14)	$\text{Weird}(\text{Ru}) \vee \text{Clown}(\text{Ru})$	Resolve (13) and (4)
(15)	$\neg \text{Clown}(\text{Ru})$	Resolve (6) and (3)
(16)	$\text{Weird}(\text{Ru})$	Resolve (14) and (15)
(17)	$\neg \text{Weird}(\text{Ru}) \vee \neg \text{Loves}(S, \text{Ru})$	From (7)
(18)	$\neg \text{Loves}(S, \text{Ru})$	Resolve (17) and (16)
(19)	$\perp$	Contradiction between (12) and (18)

Conclusion: The assumption that “Scrooge is a child” led to a *contradiction*. Therefore, by resolution:

$$\neg \text{Child}(\text{Scrooge}) \quad (\text{Scrooge is not a child.})$$